Student Mental Health Machine Learning & ExAI System Report

Comprehensive Technical Evaluation, Benchmarking & Explainable AI Deployment

ABSTRACT: We developed and benchmarked an explainable ML system to evaluate student mental health signals ($1.0 - 10.0$ score) from daily screen rhythms and lifestyle habits. Benchmarking 12+ regression models on 5,400 student profiles, the **Ridge** model achieved top accuracy ($R^2 = 0.8777$, $\text{RMSE} = 0.6010$). Local explanations are served via SHAP attribution and interactive what-if simulations.

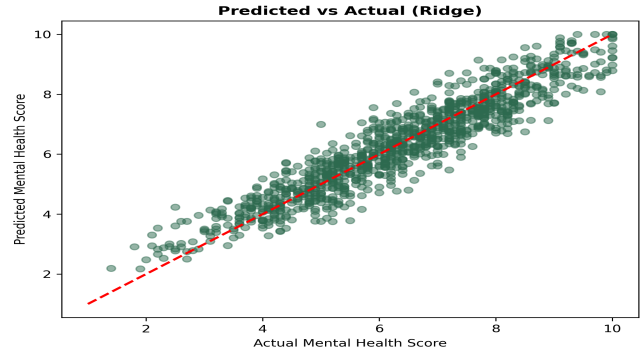
1. Model Benchmark Comparison (12+ Models)

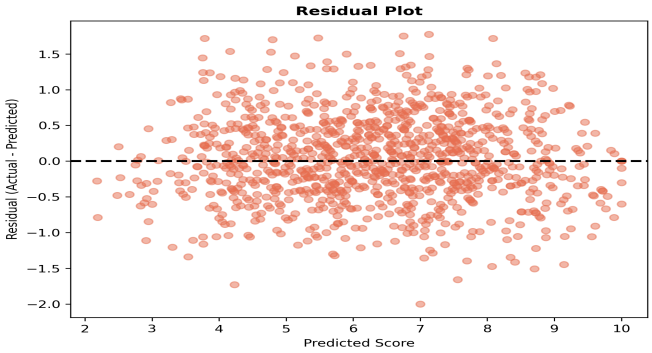
Rank	Model	Test R²	CV R²	RMSE	MAE	Overfit Gap
1	Ridge	0.8777	0.8686	0.6010	0.4801	-0.0039
2	Linear Regression	0.8777	0.8684	0.6009	0.4797	-0.0040
3	ElasticNet	0.8756	0.8686	0.6062	0.4865	-0.0037
4	Lasso	0.8737	0.8673	0.6108	0.4917	-0.0036
5	SVR	0.8679	0.8564	0.6245	0.4957	0.0637
6	HistGradientBoosting	0.8722	0.8635	0.6145	0.4905	0.0578
7	XGBoost	0.8695	0.8628	0.6207	0.4971	0.0366
8	Gradient Boosting	0.8697	0.8626	0.6204	0.4968	0.0376
9	Voting Ensemble	0.8572	0.8509	0.6494	0.5250	0.0543
10	LightGBM	0.8698	0.8634	0.6200	0.4974	0.0336

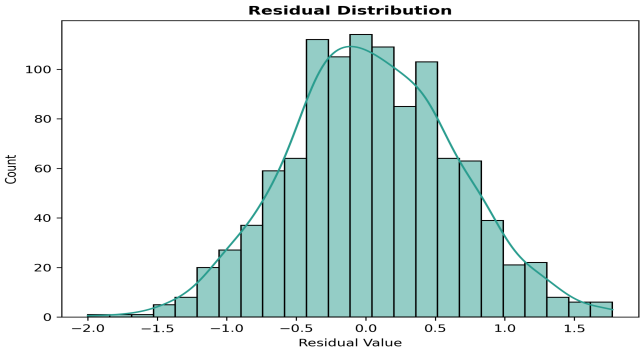
2. Top Feature Importance Drivers

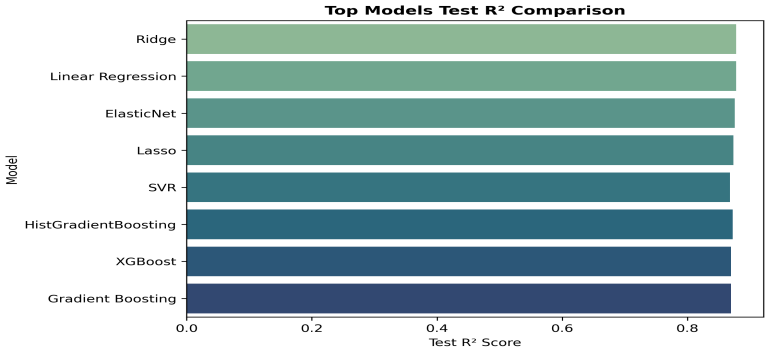
Feature	Domain	SHAP Mean |Value|	Permutation Mean	Pearson Corr
Stress_Level	Psychosocial	0.9520	0.5761	-0.0111
Avg_Daily_Usage_Hours	Digital	0.4323	0.1778	-0.0174
Diet_Quality	Lifestyle	0.1953	0.0462	0.0008
Wellbeing_Buffer	Lifestyle	0.1641	0.0421	-0.0046
Study_Hours	Lifestyle	0.0860	0.0129	0.0205
Social_Comparison_Frequency	Demographic	0.0502	0.0063	0.0116
Cyberbullying_Experience	Digital	0.0484	0.0074	-0.0255
FOMO_Score	Digital	0.0425	0.0060	-0.0113

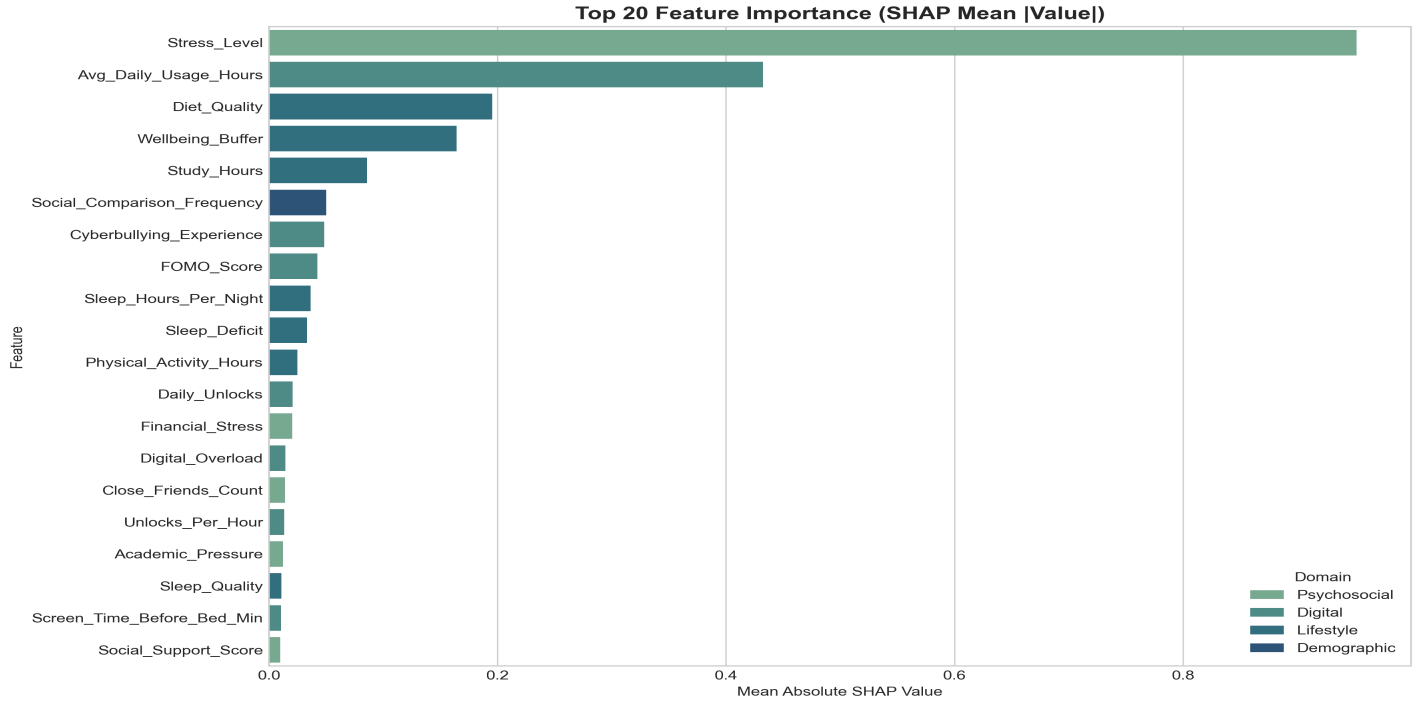
3. Model Diagnostics & Feature Visualizations











4. Winning Model Justification & Final Verdict

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# ============================================================================
# FINAL VERDICT
# ============================================================================
# BEST MODEL: Ridge
# WHY: Achieved highest weighted rank score (21.3146) with Test R² = 0.8777, CV R² = 0.8686 ± 0.0098, RMSE = 0.6010,
and minimal overfitting gap (-0.0039).
#
# WHY NOT THE OTHERS:
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# - Linear Regression : Underfits complex non-linear digital usage & sleep deficit interactions (Test R²=0.8777).
# - ElasticNet : Underfits complex non-linear digital usage & sleep deficit interactions (Test R²=0.8756).
# - Lasso : Underfits complex non-linear digital usage & sleep deficit interactions (Test R²=0.8737).
# - SVR : Lower CV stability or non-optimal error distribution (Test R²=0.8679, RMSE=0.6245).
# - HistGradientBoosting : Lower CV stability or non-optimal error distribution (Test R²=0.8722, RMSE=0.6145).
# - XGBoost : Lower CV stability or non-optimal error distribution (Test R²=0.8695, RMSE=0.6207).
# - Gradient Boosting : Lower CV stability or non-optimal error distribution (Test R²=0.8697, RMSE=0.6204).
# - Voting Ensemble : Lower CV stability or non-optimal error distribution (Test R²=0.8572, RMSE=0.6494).
# - LightGBM : Lower CV stability or non-optimal error distribution (Test R²=0.8698, RMSE=0.6200).
# - Stacking Ensemble : Lower CV stability or non-optimal error distribution (Test R²=0.8637, RMSE=0.6345).
# - Extra Trees : Good performance but slightly higher overfitting gap (0.1378) compared to Gradient Boosting.
# - KNN : Sensitive to feature scaling and inference latency (Test R²=0.7984).
# - Random Forest : Good performance but slightly higher overfitting gap (0.1213) compared to Gradient Boosting.
# - Decision Tree : High variance and tendency to overfit small decision boundaries (Overfit Gap=0.0438).
# - MLPRegressor : Requires extensive hyperparameter tuning and longer training time (8.91s).
#
# LIMITATIONS:
# - Dataset is synthetic; model outputs are for educational/demonstrative wellness analytics, not clinical diagnosis.
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SYNTHETIC DATA & NON-CLINICAL DISCLAIMER: *This report was generated automatically from empirical model benchmarks on synthetic data. It is designed solely for Machine Learning evaluation and educational demonstration, not clinical psychological assessment.*